



Graphic Era
Hill University
DEHRADUN • BHIMTAL • HALDWANI

PROJECT AND TEAM INFORMATION

Second Phase Project Report

Project Title

Patient Priority Emergency Management System

Student/Team Information

Team Name:	
Team member 1 (Team Lead) Name: Abhay Kumar Student ID: 25201300366 Email: abhay.k7417@gmail.com	

Team member 2

Name: Deepanshu Pathak

Student ID: 25201300427

Email: pathakdeepanshu74@gmail.com



Team member 3

Name: Shashank Thapliyal

Student ID: 25201300281

Email: shashankthapliyal418@gmail.com



PROJECT PROGRESS DESCRIPTION

Project Abstract

The project, titled “**Patient Priority Emergency Management System**”, aims to design and implement a console-based application using the C programming language. The primary goal of the system is to efficiently manage emergency patients by prioritizing them based on the severity of their condition rather than their arrival time.

The system utilizes an array-based Priority Queue as the core data structure to ensure that patients with critical conditions are treated first. Additionally, the project incorporates a multi-hospital allocation mechanism, where patients are assigned to suitable hospitals based on treatment availability and current patient load. Each hospital maintains its own priority queue to manage incoming patients.

The system follows a modular approach, including patient registration, priority assignment, hospital selection, queue management, treatment processing, and reporting functionalities. This project demonstrates the practical application of Data Structures and Algorithms concepts in solving real-world healthcare problems.

Updated Project Approach and Architecture

The project follows a modular and Data Structures-oriented **approach** using the C programming language in a console-based environment. The system is designed to simulate a real-world emergency patient management scenario by integrating priority-based processing and hospital allocation logic.

The core of the system is an array-based Priority Queue, which is implemented separately for each hospital. Patients are inserted into the queue based on their priority level (severity of condition) and arrival time, ensuring that critical patients are treated first. The system uses structures and arrays to represent patients and hospitals efficiently.

The architecture consists of multiple modules, including patient registration, priority assignment, hospital selection, queue management, treatment processing, and reporting.

Tasks Completed

Task Completed	Team Member
1. Problem analysis and requirement identification	Abhay Kumar
2. Design of system architecture and module structure	Deepanshu Pathak
3. Creation of Patient and Hospital data structures	Deepanshu Pathak
4. Implementation of array-based Priority Queue (insertion & deletion)	Deepanshu Pathak
5. Development of priority assignment logic based on severity	Abhay Kumar
6. Implementation of hospital allocation algorithm	Abhay Kumar
7. Menu-driven console interface design	Shashank Thapliyal
8. Basic testing of priority queue operations	Shashank Thapliyal

Challenges/Roadblocks

1. Difficulty in implementing an efficient array-based Priority Queue, especially maintaining correct order during insertion and deletion
2. Handling tie-breaking using arrival time when multiple patients have the same priority
3. Designing the hospital allocation algorithm based on treatment availability and patient load
4. Managing multiple hospitals, each having its own priority queue, increased system complexity
5. Debugging logical errors and handling C programming challenges such as memory management and indexing
6. Ensuring proper modular structure for smooth integration of all components

Tasks Pending

Task Pending	Team Member (to complete the task)
1. Implementation of search functionality for patients 2. Development of report generation module 3. Integration of all modules into final system 4. Testing with multiple scenarios and edge cases	Abhay Kumar Deepanshu Pathak Abhay Kumar Shashank Thapliyal

Project Outcome/Deliverables

1. A fully functional console-based application developed using C programming language
 2. Implementation of array-based Priority Queue for efficient patient management
 3. Priority-based treatment system ensuring critical patients are handled first
 4. Multi-hospital allocation system based on treatment availability and patient load
- Menu-driven interface for smooth user interaction and system control
5. Modular implementation including registration, allocation, queue management, treatment, and reporting
 6. Algorithm design and time complexity analysis of core operations
 7. System architecture and flow diagrams representing complete system design
- Well-structured source code and project documentation

Progress Overview

The project has made significant progress and most of the core components have been successfully implemented. The design of the system architecture, data structures, and priority queue logic has been completed. The implementation of the array-based priority queue and hospital allocation mechanism is functioning correctly.

Currently, the project is in the integration and testing phase, where individual modules are being combined and tested with different scenarios. Basic functionalities such as patient registration, priority assignment, and queue operations are already working as expected.

Some features, including report generation, advanced testing, and final optimization, are still in progress. Overall, approximately 70–80% of the project has been completed, and the remaining work mainly involves refinement, testing, and documentation.

Codebase Information

The project codebase is currently maintained locally and developed using the C programming language in a modular structure. Each module, including patient registration, priority queue operations, hospital allocation, and reporting, is implemented as separate functions to ensure clarity and maintainability.

The code is organized into logical sections, making it easy to understand and modify. Regular updates and improvements have been made during development to enhance functionality and fix logical errors. Version control is managed manually by maintaining updated copies of the source code.

GITHUB LINK: <https://github.com/Abhaykumar39/Patient-Emergency-Priority-System>

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
1. Priority Queue Insertion Test	Pass	Patients inserted correctly based on priority
2. Priority Queue Deletion Test	Pass	Highest priority patient removed correctly
3. Hospital Allocation Test	Pass	Patients assigned to correct hospital based on treatment
4. Tie-breaking (Same Priority) Test	Pass	Arrival time handled correctly
5. Search Functionality Test	In-Progress	Under implementation

Deliverables Progress

Console-based C application → In Progress

Array-based Priority Queue implementation → Completed

Hospital allocation logic → Completed

Menu-driven interface → Completed

Search functionality → In Progress

Report generation module → In Progress

Testing and validation → In Progress

Documentation and report → In Progress